HOW TO BE A SOC ANALYST IN 2026

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INTRODUCTION

As our daily lives become increasingly connected to the digital world, ensuring its security has never been more important. There are threat actors ranging from kids looking to cause mischief to nation states bent on creating chaos for their adversaries, and the battleground is ones and zeros traveling through cyber space.

But the attackers are not the only ones who operate in the digital aether. Defenders monitor and protect the networks and information that keep the world as we know it running smoothly. One of these defenders is the SOC Analyst.

This guide will explore the duties of a SOC analyst, some lasting motivations for pursuing defensive cybersecurity, and resources for getting started and progressing in the role.

WHY BECOME A SOC ANALYST?

Understanding your motivation is a beneficial move, no matter the pursuit. So, asking yourself “Why do I want to become a SOC Analyst?” is an important first step.

Defensive cybersecurity is a rewarding and usually well-paying job, but it can also be repetitive, high-pressure, and demanding. These factors are a recipe for burnout from the profession and possibly the industry.

There are a few attributes that can help you through those moments when you question why you thought that being a SOC Analyst would be a good idea.

Service

You are helping people. Even if you are protecting the network of the biggest corporation, at the end of the day, threat actors are trying to get customer information or disrupt life as we know it. If the trains don’t run on time (or the power doesn’t work), there is a trickle-down effect that has outsized implications for lots of people.

Justice

Similar to service, if you want to see the bad guys lose, defensive cybersecurity can scratch that itch. Identifying a malicious IP or file and locking a threat actor out of the system can be an exhilarating feeling, and can provide a little sharpness to contrast the more tedious days.

Curiosity

This is a vital trait for anyone in the field of cybersecurity, red or blue. So many times, that extra ounce of “why?” or “how?” can mean the difference between uncovering the problem or not. If chewing on a problem for long periods of time before finding the answer brings you joy, then a SOC Analyst role might be for you.

These are just a few examples of answers to that basic ‘why’ question. You may find another, but you would be doing yourself a favor by giving it some thought before you invest your time and money.

WHY SOC ANALYSTS MATTER

The purpose of cyber defenders is not just to protect networks from intrusion. It is to instill and maintain trust. As the wise matriarch of the Heeler family once said, “The whole point of promises is to build trust. If there’s no trust, none of this is possible… No libraries, no roads, no power lines.” When people hand over their personal data or financial information to an organization, they are trusting that organization to keep that data safe and not abuse or lose it. If an organization breaks that promise, people are less likely to trust another organization in the future. That could mean a small business that wants to expand beyond its local market may have a hard time convincing people to entrust it with their credit card information.

Beyond this, institutions like public utilities are counted on by the whole world to deliver clean water, consistent power, and consumer goods. The ripple effects of some attacks extend far beyond the initial target and intent (see NotPetya), and many times these attacks could have been prevented in some organizations through the most basic cybersecurity hygiene.

It may be frustrating at times (especially in bigger organizations where the pace of adoption for cybersecurity changes can be slow) to see that measures like MFA on company accounts are not implemented as quickly as you'd like. But it's important to remember that even 99.9% adoption for such security controls still leaves the possibility of a breach.

This is why the work of the SOC analyst is so important. Whether through defensible means, like misconfigurations, or indefensible, such as zero-day exploits, proper monitoring and logging are the next line of defense and means by which the attack is quickly detected and the damage mitigated. It is truly noble work.

WHAT DOES A SOC ANALYST DO?

In the range of responsibilities that make up the defensive cybersecurity field, the SOC Analyst is primarily a monitor. When someone is knocking on the door of the network, the SOC Analyst is the one looking through the peephole to see who's there. When the trip wires inside the network are disturbed, the SOC Analyst is the first one on the scene.

In more technical terms, the SOC Analyst receives alerts from the organization's detection systems and investigates whether or not these situations are malicious. This can be as simple as triaging the alerts and flagging the suspicious ones to be investigated, and disregarding the rest, or taking initial steps to limit access for possible threat actors, such as blocking IP addresses or revoking user access.

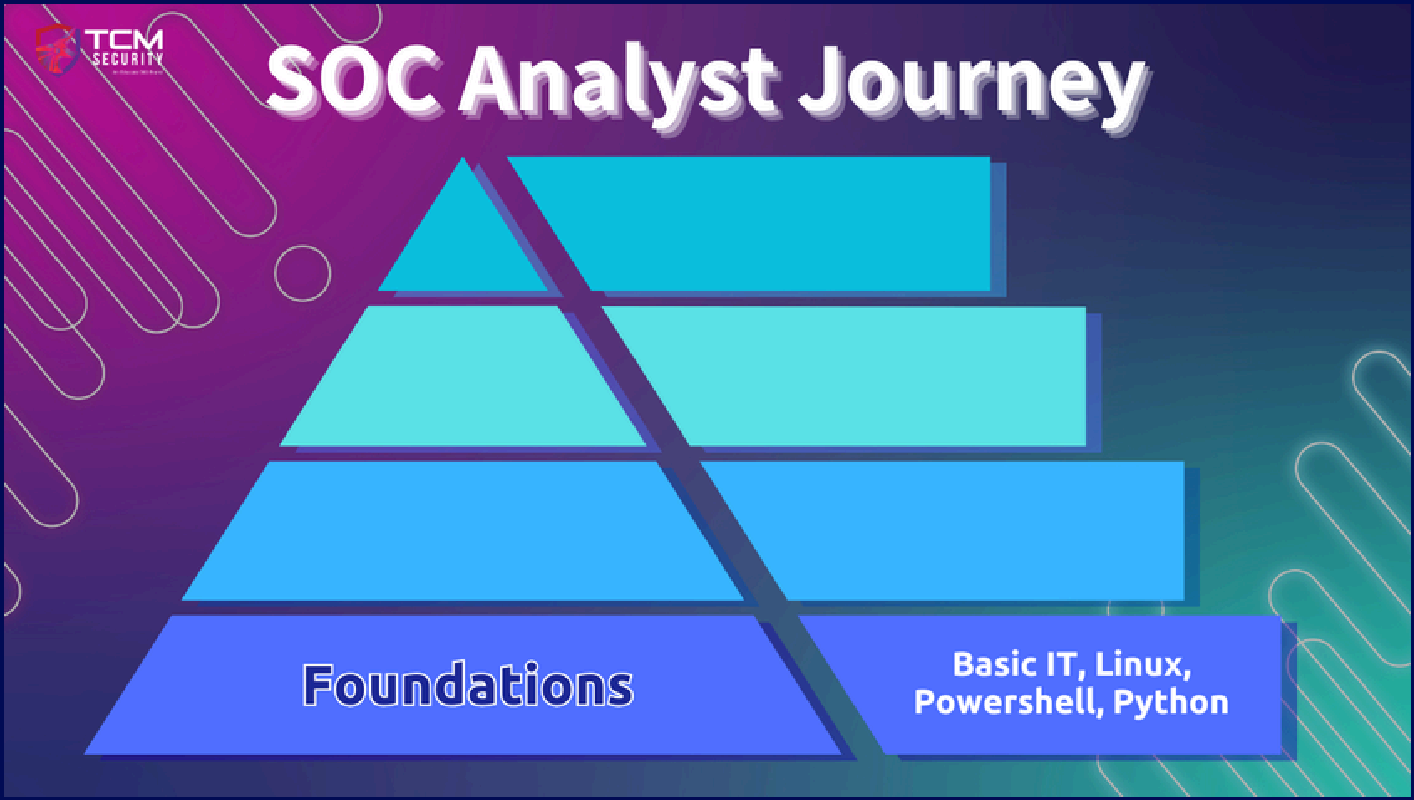
Here is a general overview of the blue team ecosystem:

- **Monitoring:** Manning the castle walls and watching for signs of attack.
- **Incident Response:** Kicking the bad guys out and restoring order.
- **Digital Forensics:** Discovering how intruders got in and what they did.
- **Threat Hunting:** Finding threats before they strike.
- **Defensive Engineering**: Building better defenses and smarter detection tools.
- **Compliance**: Making sure the walls are up to code

There can be some overlap in these roles, but these make up the general foundations of a good defensive cybersecurity operation. SOC Analysts may wear one or more of these hats, become a liaison between these departments, or expand into other roles during their career, so knowing they exist is a good starting point.

GETTING STARTED AS A SOC ANALYST

Cybersecurity is all about understanding IT infrastructure, because how can you defend what you don't understand? Basic networking and being familiar with the environments and languages that are used in IT are going to be the backbone of any successful cybersecurity career.



FOUNDATIONAL SKILLS

There are tons of resources for studying these concepts, and many affordable certifications that can signal competence to a hiring manager, but really getting your hands dirty with labs and real-world experience will help establish that foundation. Help a friend set up a home network. Set up and segment virtual networks in a safe environment. There are several ways to practice these skills in the real world.

Here are some resources that will provide some excellent training for getting started and preparing for certifications:

1) BASIC IT SKILLS

Practical Help Desk – TCM Security Academy (FREE)

The 19-hour **Practical Help Desk** course by TCM Security Academy is a free, hands-on program designed to prepare students for entry-level IT roles. It covers essential skills needed to excel at a help desk position, including troubleshooting common technical issues, managing tickets, and customer service fundamentals.

The course emphasizes practical, real-world scenarios to build confidence in resolving hardware, software, and networking challenges. Ideal for beginners, it offers a straightforward path to building foundational IT knowledge and experience, making it an excellent starting point for those pursuing a career in tech.

220-1101 and 220-1102 A+ Courses – Professor Messer (FREE)

Professor Messer's **220-1101** and **220-1102** A+ courses cover essential knowledge needed for passing the CompTIA A+ certification, focusing on both hardware and software fundamentals.

- The 220-1101 (Core 1) course dives into hardware technologies such as networking devices, cables, and peripherals, along with virtualization and mobile device management. It emphasizes practical troubleshooting, from understanding network configurations to managing hardware components like motherboards and storage systems.

- The 220-1102 (Core 2) course shifts focus to operating systems, security, and software troubleshooting. It includes modules on Windows, Linux, and macOS features, explores physical and logical security best practices, and provides strategies for tackling malware, social engineering, and mobile device security. Core 2 also highlights practical IT skills like Active Directory management and securing SOHO networks.

Jason Dion – A+ Core 1 and 2 (PAID)

If you are pursuing the CompTIA A+ certification, Jason Dion's courses are also highly recommended. They cover many of the same topics covered in Professor Messer's courses, but the main difference is that they come with practice quizzes and exams that will help you prepare for the style of the certification exam.

- https://www.udemy.com/course/comptia-a-core-1/
- https://www.udemy.com/course/comptia-a-core-2/

2) LINUX SKILLS

Linux 100: Fundamentals – TCM Security Academy (FREE)

The free **Linux Fundamentals** course introduces essential Linux concepts, including file management, permissions, and basic scripting. It's a great starting point for beginners wanting a structured introduction to the operating system.

Linux 101 – TCM Security Academy (PAID)

For those seeking deeper, structured learning, TCM Security Academy offers a **Linux 101 course,** which builds upon the Linux 100 course mentioned above. This course covers the foundations needed to become comfortable using Linux, with practical exercises that prepare you for real-world scenarios.

Bandit - OverTheWire (FREE)

Bandit is a fantastic series of challenges designed to teach you Linux through practical problem-solving, helping you build both knowledge and troubleshooting skills.

3) WINDOWS POWERSHELL

Don Jones – Learn Windows PowerShell in a Month of Lunches (FREE)

This **playlist** offers a clear, approachable introduction to PowerShell, guiding beginners through essential concepts in short, digestible lessons. It walks you through the fundamentals—like navigating the shell, running commands, and understanding providers—to practical skills such as scripting, automation, and real-world administrative tasks.

Fernando Tomlinson – UnderTheWire, Wargames (FREE)

Like Bandit mentioned above, **UnderTheWire** is a PowerShell-based "wargame" / training platform that lets you practice and hone your PowerShell, Windows-admin, and security skills through progressively harder interactive challenges.

PowerShell for Hackers (FREE)

PowerShell for Hackers is a resource hub combining: PowerShell-related hacking/penetration-testing scripts and tools, tutorials or blog-style writeups, and interactive challenges to practice PowerShell skills in a security-oriented context.

Intro to PowerShell Series - TCM Security (FREE)

We also have a free series on our **YouTube channel** designed to introduce newbies to PowerShell in a defensive security context.

4) PYTHON

Programming 100: Fundamentals – TCM Security Academy (FREE)

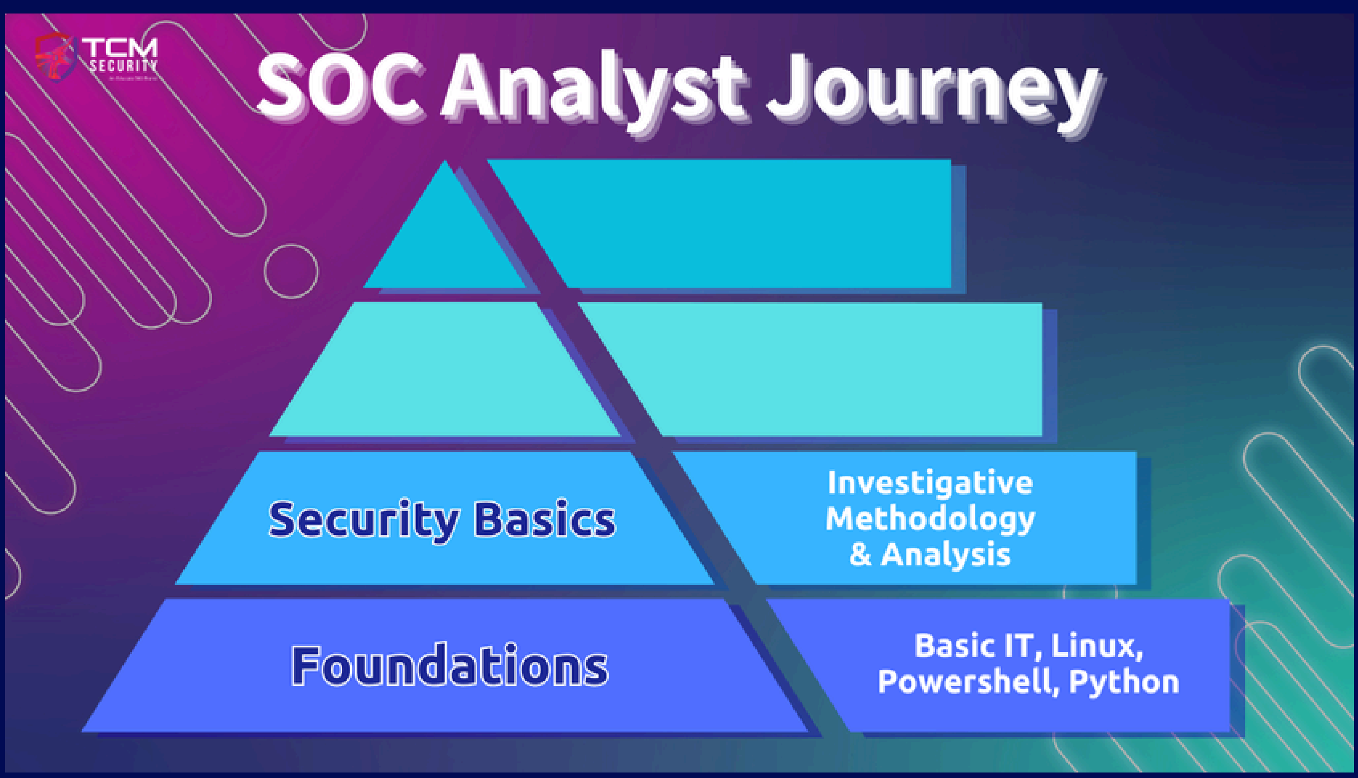
For those completely new to programming, **Programming 100 Fundamentals** offers a beginner-friendly introduction. This course covers the basics of coding with Python, including variables, loops, and control structures, providing a solid foundation for further programming studies.

30 Days of Python (FREE)

30 Days Of Python is a free, open-source “course / challenge” designed to teach (or brush up on) the Python programming language in a structured, day-by-day format.

SECURITY BASICS

Getting your feet wet with IT work will take you far, but there are some fundamentals of cybersecurity to acquire if you want to be prepared for a SOC Analyst role. Studying for some entry-level certifications will fill many of the gaps coming straight out of IT, and hands-on practice executing methodologies and using tools will be a real benefit.



Developing an investigative methodology is going to be a crucial step in pursuing defensive cybersecurity. A natural curiosity and desire to unravel mysteries are helpful traits, but turning those into a scientific method for finding evil in a network is what will get the job done. Learning frameworks like **MITRE ATT&CK** are a good starting point for understanding how threat actors conduct recon, prep, and execute attacks, and otherwise disrupt networks. There are several resources for gaining this knowledge and practice:

Practical Security Fundamentals – TCM Security Academy (FREE)

Practical Security Fundamentals introduces students to general security concepts and operations, as well as the common threats, vulnerabilities, and mitigations faced in the field. This course sets students up to pursue more advanced topics.

SY0-701 Security+ Course – Professor Messer (FREE)

Professor Messer offers a comprehensive **Security+ video series** covering all exam objectives, including topics like network security, incident response, and access control.

Security Operations (SOC) 101 – TCM Security Academy (PAID)

The 30-hour **SOC 101** course offers a detailed introduction to Security Operations Centers (SOCs) and the role of a SOC Analyst. It covers core topics such as log analysis, incident response, and monitoring tools, providing practical skills to excel in entry-level security roles. Ideal for those pursuing a career as a SOC Analyst by learning how to defend, this course bridges the gap between theoretical knowledge and real-world operations.

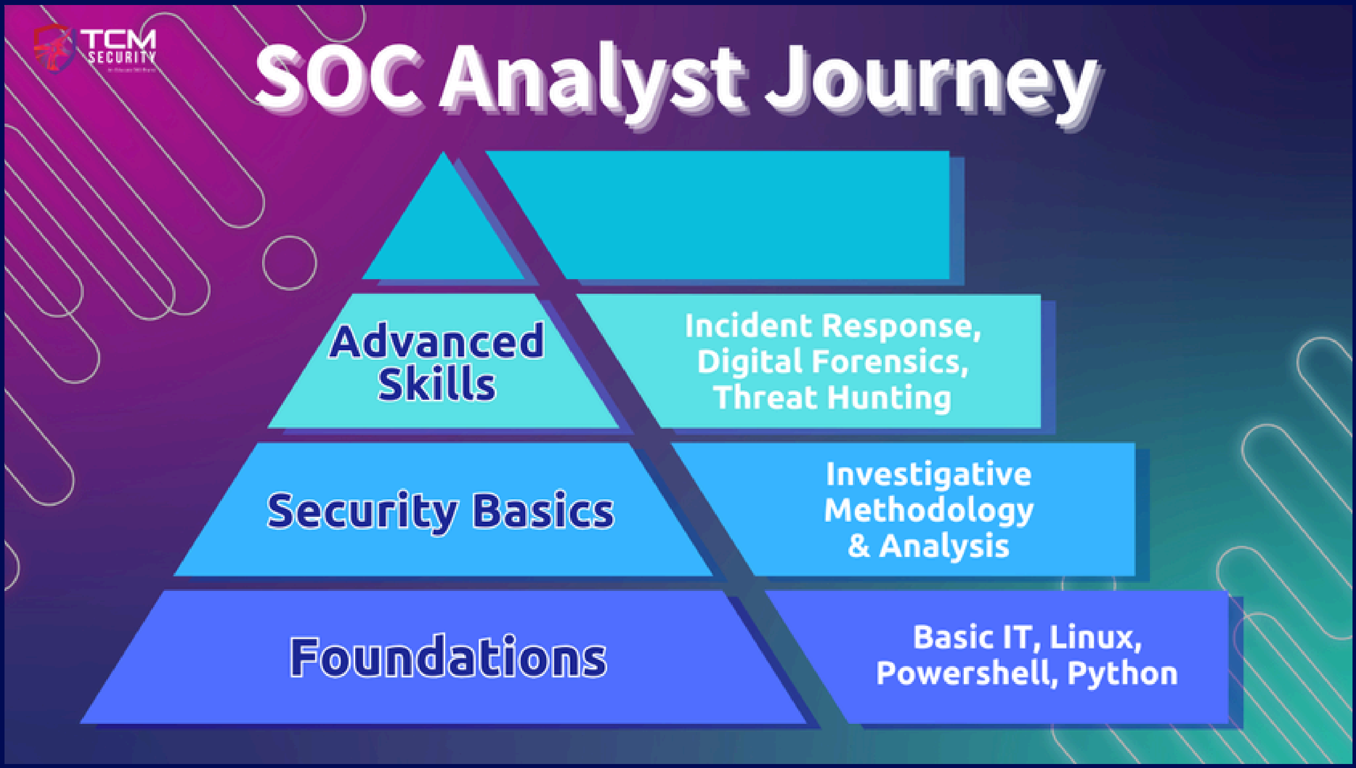
Practical SOC Analyst Associate (PSAA) - TCM Security (PAID)

The **PSAA certification** builds on the course material of SOC 101. It is an associate-level security operations and incident response exam that will test your ability to use analysis tools, interpret artifacts, and apply investigation methodologies to respond to incidents at a Tier 1 and Tier 2 level.

ADVANCED SECURITY SKILLS

Once you have some time working in a SOC and are racking up experience, you may start to wonder what advancing in the field looks like. There are many disciplines that you could choose to pursue, including incident response, digital forensics, threat hunting, and malware analysis, among others.

Just as the path from IT to SOC required developing skills and then building on those with education, training, and practice, moving into one of these advanced disciplines will build on the experience you gain in the SOC. If possible, it is a good idea to look for ways to incorporate the responsibilities attached to these skills into your daily duties.



1) INCIDENT RESPONSE & THREAT HUNTING

Security Operations (SOC) 201 – TCM Security Academy (PAID)

The 25+hour **SOC 201** course teaches analysts how to identify, hunt, and respond to real-world adversary tactics and techniques. With a practical, hands-on focus, the curriculum provides realistic scenarios where you will investigate sophisticated threats across multiple systems, learning to detect and respond effectively in enterprise-scale environments. The course also integrates proactive threat hunting as part of a continuous detection and response cycle, giving you the mental models to identify active threats, uncover gaps, and feed insights back into investigative processes to improve future detection and response efforts.

Practical SOC Analyst Professional (PSAP) - TCM Security (PAID)

The **PSAP certification** tests your mastery of the SOC 201 material. It is an intermediate-level threat hunting and incident response exam that assesses your ability to proactively identify intrusions, reconstruct attacker activity, analyze evidence, and develop actionable recommendations to contain, eradicate, and recover from a realistic compromise.

Threat Hunting Training Course – Active Countermeasures (FREE)

The **Active Countermeasures Threat Hunting Training** is a free, six-hour Level-1 course that teaches you how to identify malicious activity by analyzing both network and host data across desktops, servers, and other connected systems. It includes realistic hands-on labs using pre-configured virtual machines with real command-and-control traffic, allowing you to practice detection techniques in a safe environment.

The Threat Hunting Project (FREE)

The **Threat Hunting Project** is a community-driven and open-source "threat hunting" repository / knowledge base. It collects and organizes publicly available threat-hunting techniques, procedures, research, and resources into a structured library so that both beginners and experienced defenders can use and contribute.

Detection Engineering for Beginners - TCM Security Academy (PAID)

This detection engineering course combines theory with hands-on detection engineering practice. In it, you will build a home lab and run through three different attack scenarios, learning how to document our detections and automate processes along the way.

2) DIGITAL FORENSICS

Introduction to Memory Forensics– 13Cubed (FREE)

The **video** reviews the importance of memory forensics to uncover evidence of malicious activity, malware, or indicators of compromise that may not appear on disk. The video also walks through how memory forensics fits into a broader digital forensics / incident response workflow, especially in scenarios where attackers use advanced techniques.

Introduction to Windows Forensics - 13Cubed (FREE)

This **video** is an introductory lecture on forensic analysis of Windows systems. It reviews the kinds of artifacts and data sources available on Windows machines, including both live/volatile and persistent/non-volatile. It will teach you how to leverage Windows-specific artifacts to reconstruct system and user activity.

Practical Windows Forensics - TCM Security Academy (PAID)

In the **Practical Windows Forensics** course you'll complete a full digital forensic investigation of a Windows system. You'll run an attack simulation scripts on the victim VM that simulates attack patterns as commonly observed by threat actors in the industry to create a realistic setting for our investigation. From there, you'll perform a forensic investigation, beginning with the data collection, examination and extraction before diving deeper into the analysis of the information at hand.

Digital Forensics and Incident Response - SANS (FREE)

The **SANS DFIR channel** offers a broad, professionally curated collection of videos on digital forensics, incident response, threat hunting, and cyber-threat intelligence ranging from beginner-friendly "101" intros to advanced, real-world case studies and forensic techniques.

3) MALWARE ANALYSIS

Introduction to Malware Analysis - 13Cubed (FREE)

In this **video**, you will compare an unpacked and packed version of a very basic Windows binary and note the major differences. Then, pretending the packed binary is malware, you'll perform dynamic analysis on the file using x64dbg, with the goal of allowing the code to execute until the binary unpacks itself in memory. Once unpacked, you'll explore how to dump that binary to disk for further analysis.

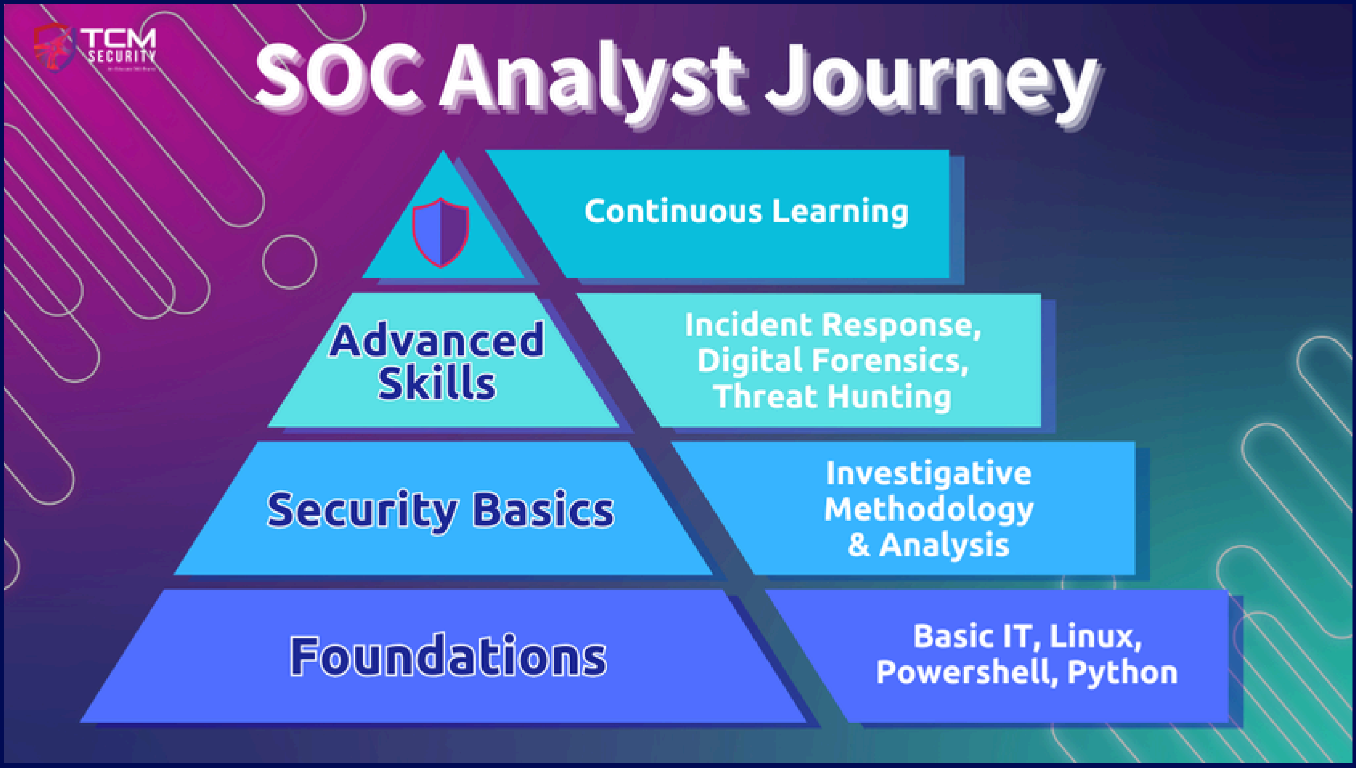
Practical Malware Analysis & Triage - TCM Security Academy (PAID)

In this **course**, you will get hands-on experience with malware. First, you will learn to handle malware safely and construct an isolated lab environment. Then, you will learn the basics of malware analysis on samples designed to teach you the core analysis concepts. As the labs progress, the level of offensive tradecraft employed by these samples grows. By the end of the course, you'll be using automated workflows and advanced analysis to extract key facts about real-world specimens.

BUILDING YOUR CAREER

Learning never stops, so don't let imposter syndrome prevent you from applying to jobs and starting your career. In addition to the technical skills you need to learn, there are other soft skills and experiences that can help increase your chances of landing a role.

While there's no one magic word you can say or thing you can do to summon your dream role, making the right connections can drastically increase your chances of success. Let's look at a few ways you can become a more competitive job applicant.



NETWORKING

Any conversation about breaking into, or advancing in, a field should include a section on professional networking. While this is not a technical advancement, it is nonetheless one of the most important factors in having your application reviewed and securing interviews for positions. At least some of your professional development time should be spent meeting people in your industry and maintaining those relationships in lieu of purely technical practice.

Here are a few links to check out if you're interested in upping your people networking game.

- **Soft Skills for the Job Market**: https://academy.tcm-sec.com/p/soft-skills-for-the-job-market
- **Networking on Social Media**: https://tcm-sec.com/social-media-cybersecurity-networking/
- **Interview Prep:** https://tcm-sec.com/how-to-prepare-for-your-cybersecurity-interview/
- **Resume Writing:** https://tcm-sec.com/crafting-a-winning-cybersecurity-resume-8-tips-and-tricks/

CONCLUSION

The cybersecurity job market ebbs and flows, but time spent developing skills and getting to know people in the industry is never wasted. Learning the basics, developing skills, and learning continuously are steps that will give you the know-how to succeed as a SOC analyst. Sharpening your soft skills and networking with peers and industry professionals will help take you the rest of the way into a position. It may take some patience and persistence, but those are also learned skills and hallmarks of a good SOC analyst.

Good luck out there.

ALL OF THE LINKS!

FOUNDATIONAL SKILLS

- Practical Help Desk - TCM Security Academy (FREE)
- 220-1101 (Core 1) - Professor Messer (FREE)
- 220-1102 (Core 2) - Professor Messer (FREE)
- A+ Core 1 - Jason Dion (PAID)
- A+ Core 2 - Jason Dion (PAID)
- CompTIA A+ (PAID)
- Linux 100: Fundamentals - TCM Security Academy (FREE)
- Bandit Wargame - OverTheWire (FREE)
- Linux 101 - TCM Security Academy (PAID)
- Programming 100: Fundamentals - TCM Security Academy (FREE)
- 30 Days of Python (FREE)
- Learn Windows PowerShell in a Month of Lunches - Don Jones (FREE)
- UnderTheWire, Wargames - Fernando Tomlinson (FREE)
- PowerShell for Hackers (FREE)
- Intro to PowerShell Series - TCM Security (FREE)

BASIC SECURITY SKILLS

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- Security Operations (SOC) 101 - TCM Security Academy (PAID)
- Practical SOC Analyst Associate - TCM Security (PAID)

ADVANCED SECURITY SKILLS

- Security Operations (SOC) 201 - TCM Security Academy (PAID)
- Practical SOC Analyst Professional - TCM Security (PAID)
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- The Threat Hunting Project (FREE)
- Detection Engineering for Beginners - TCM Security Academy (PAID)
- Introduction to Memory Forensics - 13Cubed (FREE)
- Introduction to Windows Forensics - 13Cubed (FREE)
- Practical Windows Forensics - TCM Security Academy (PAID)
- Digital Forensics and Incident Response - SANS (FREE)
- Introduction to Malware Analysis - 13Cubed (FREE)
- Practical Malware Analysis & Triage - TCM Security Academy (PAID)

NETWORKING

- Soft Skills for the Job Market: https://academy.tcm-sec.com/p/soft-skills-for-the-job-market
- Networking on Social Media: https://tcm-sec.com/social-media-cybersecurity-networking/
- Perseverance: https://tcm-sec.com/perseverance-cybersecurity-job-search/
- Interview Prep: https://tcm-sec.com/how-to-prepare-for-your-cybersecurity-interview/
- Resume Writing: https://tcm-sec.com/crafting-a-winning-cybersecurity-resume-8-tips-and-tricks/